M5STACK

StickS3

1. Description

StickS3 is a compact and high-performance programmable controller specifically designed for remote control and IoT applications. At its core, it features the ESP32-S3-PICO-1-N8R8 main control chip, supports 2.4 GHz Wi-Fi wireless communication, and comes with 8MB Flash and 8MB PSRAM, meeting the needs of diverse application development while delivering outstanding performance and expandability. For human-machine interaction, it is equipped with a 1.14" LCD display, a 6-axis IMU sensor, and programmable buttons. The audio system employs the ES8311 mono codec, combined with a high-sensitivity MEMS microphone and an AW8737 power amplifier, enabling clear sound pickup and high-fidelity audio output for voice recognition and interactive experiences. Additionally, it integrates IR transmitter and receiver tubes, along with a 250mAh lithium battery, making it suitable for applications such as smart home control, AI voice assistants, and IoT project development.

2. Specifications

Specification	Parameters
SoC	ESP32-S3-PICO-1-N8R8 @ Dual-core Xtensa LX7 processor, up to 240MHz main frequency
PSRAM	8MB
Flash	8MB
Input Power	USB: DC 5V
Audio CodeC	ES8311: 24-bit resolution, using I2S protocol
MEMS Microphone	MSM381A3729H9BPC, Signal-to-Noise Ratio (SNR): ⩾65 dB
Speaker	AW8737 power amplifier,2011 cavity speaker: 1W@8Ω
Operating Temperature	0 ~ 40°C

3. Quick Start

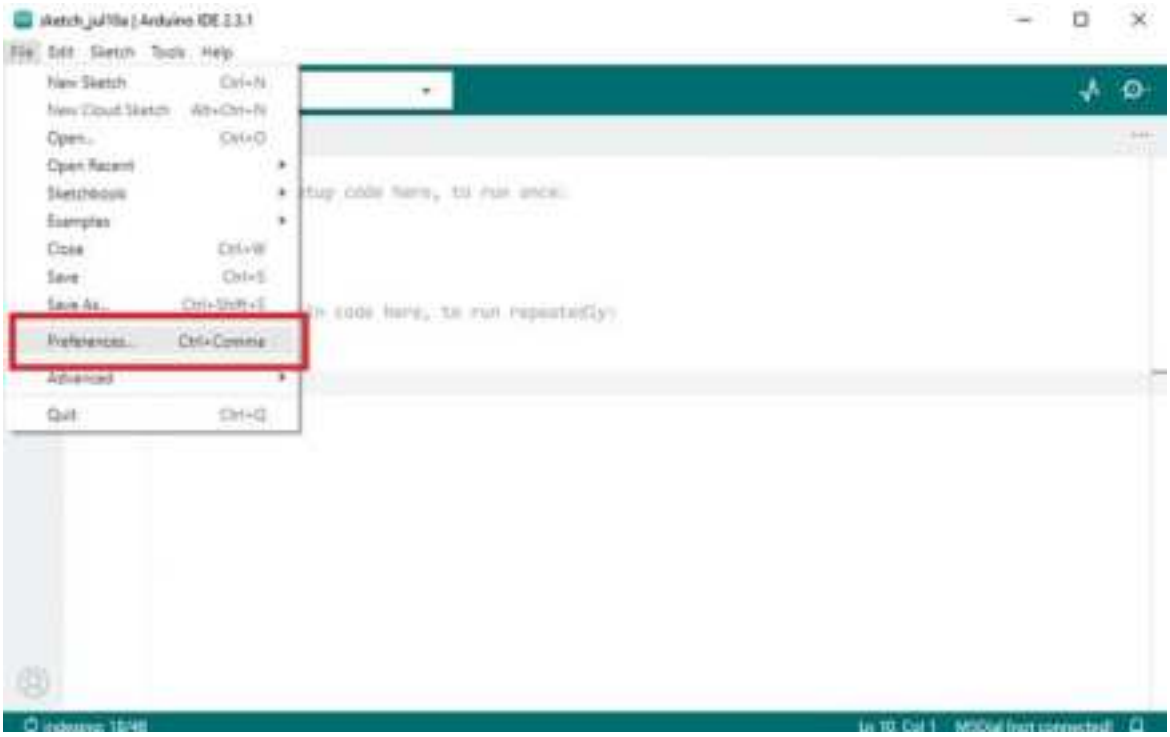
3.1 Preparation

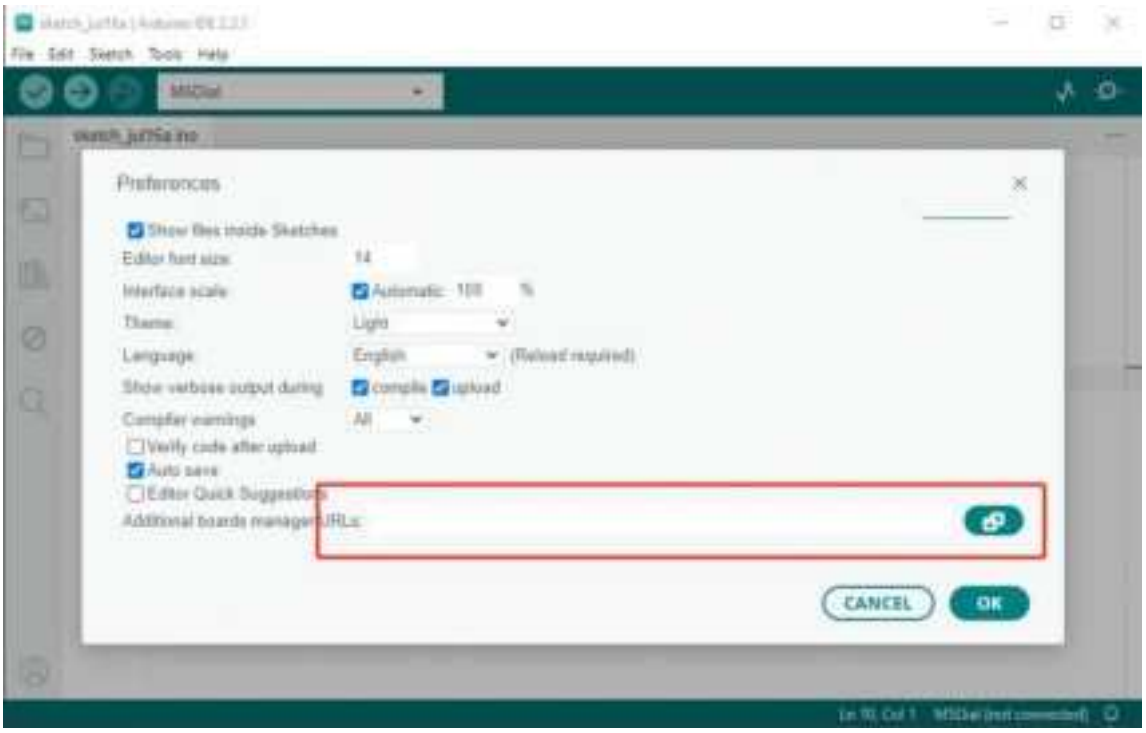
1. Visit the official Arduino website and install the Arduino IDE

https://www.arduino.cc/en/Main/Software

2. Add the following Board Manager URL to File → Preferences → Additional Boards Manager URLs:

https://espressif.github.io/arduino-esp32/package_esp32_dev_index.json

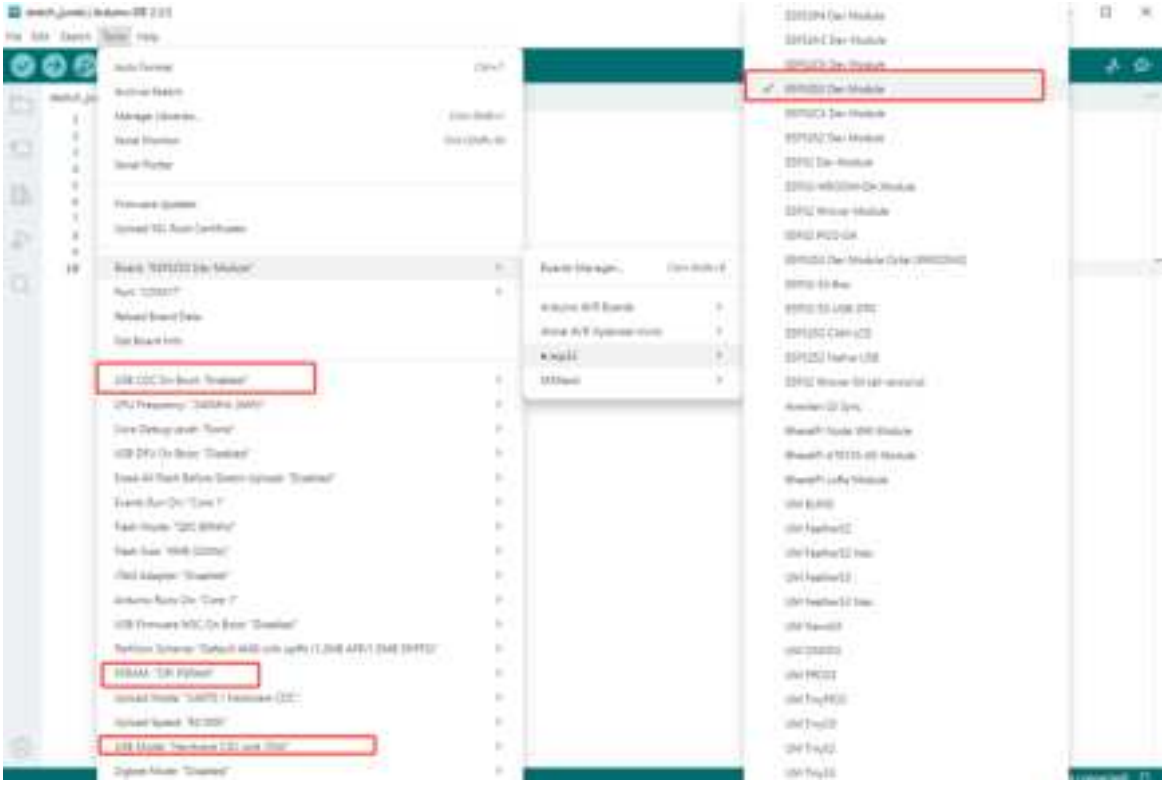




3. Open the Boards Manager, search for "ESP32", and click install.

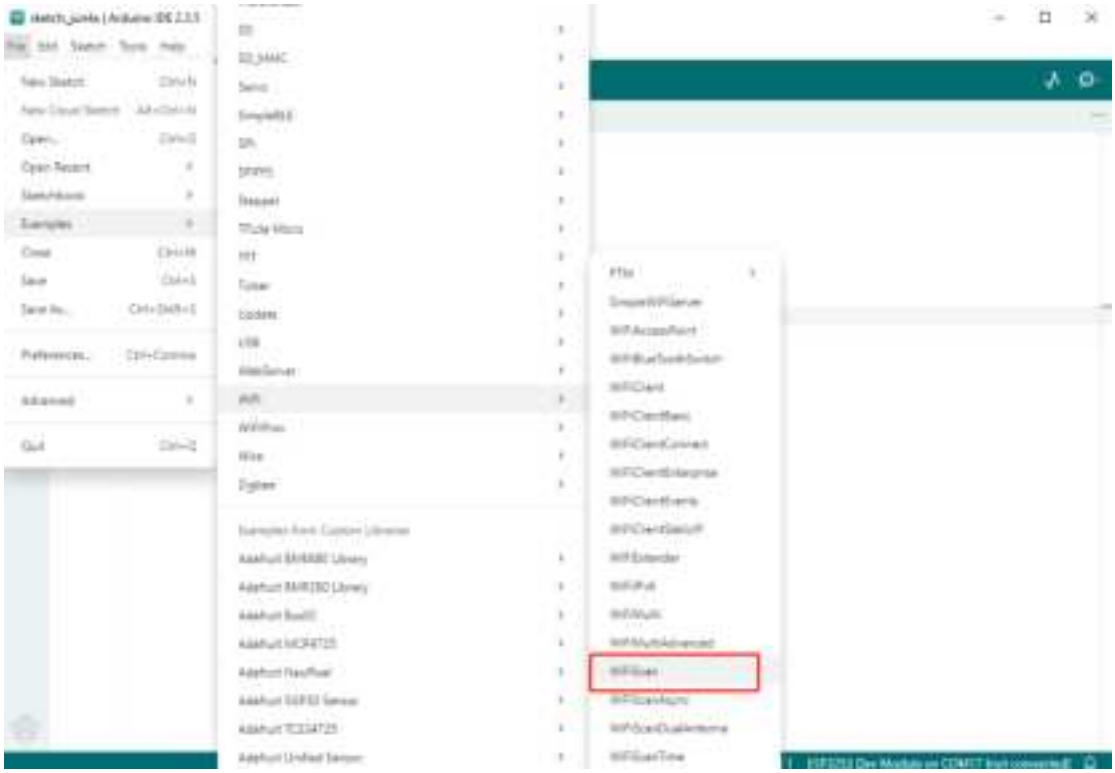
4. After installation, select the board "ESP32S3 Dev Module"

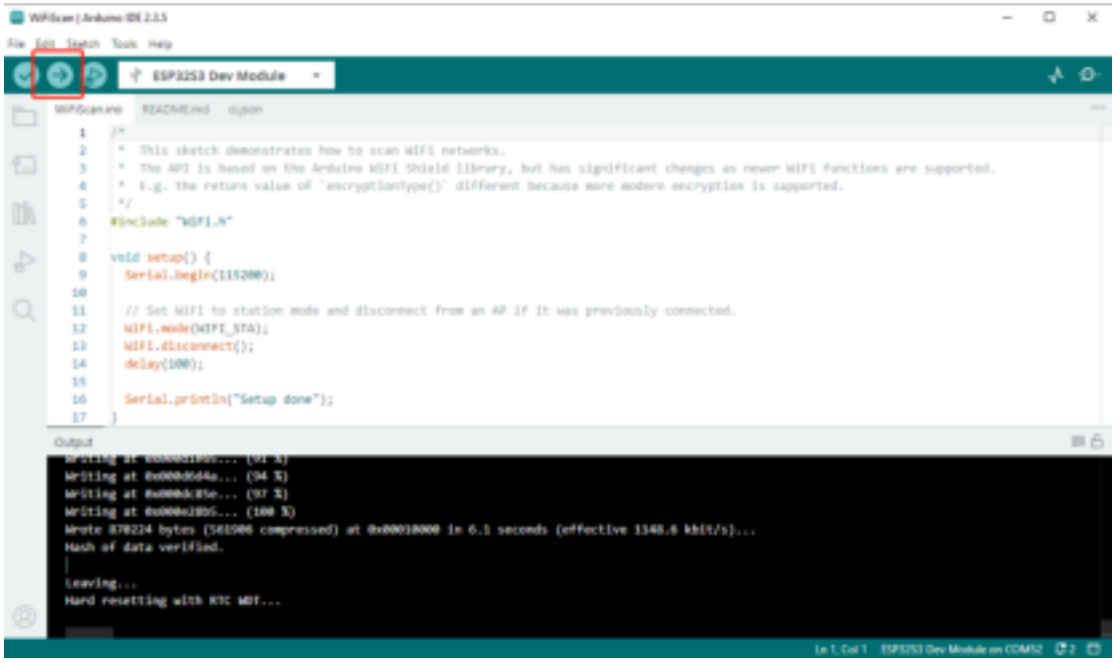
5. Configure the following options. USB CDC On Boot: "Enabled", PSRAM: "OPI PSRAM", USB Mode: "Hardware CDC and JTAG"



3.2 Wi-Fi Scan

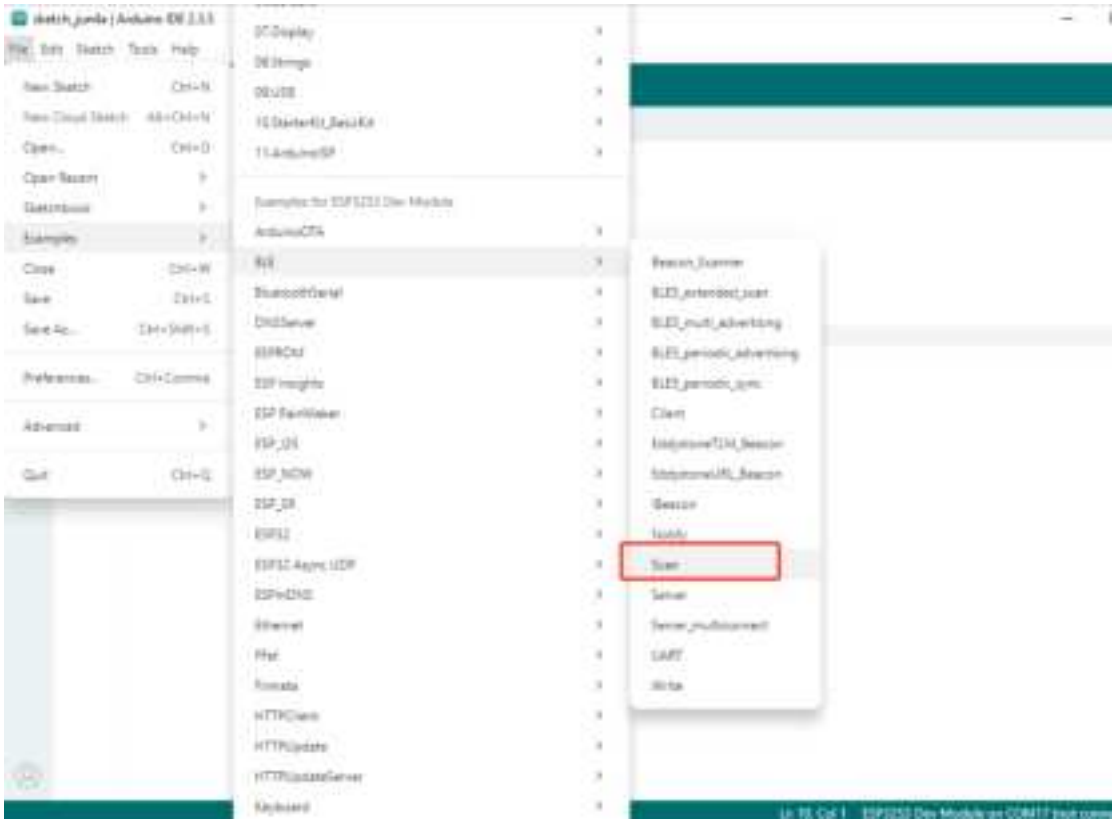
Select the example program "Examples" → "WiFi" → "WiFiScan", choose the port corresponding to your device, and click the compile and upload button in the top-left corner.
After uploading is complete, open the Serial Monitor to view Wi-Fi scan information.

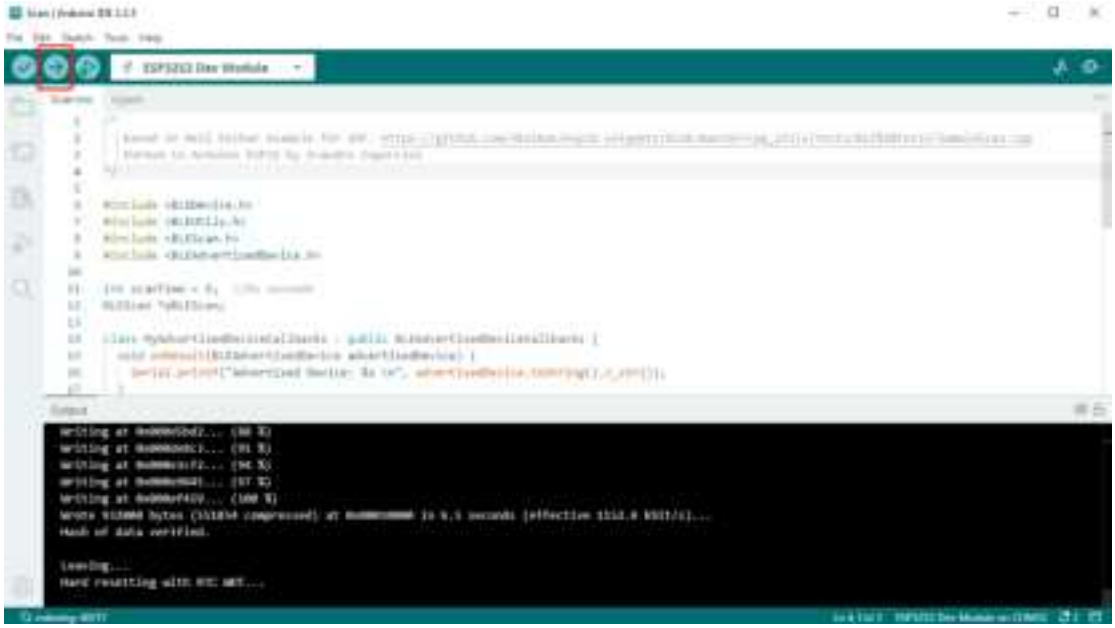


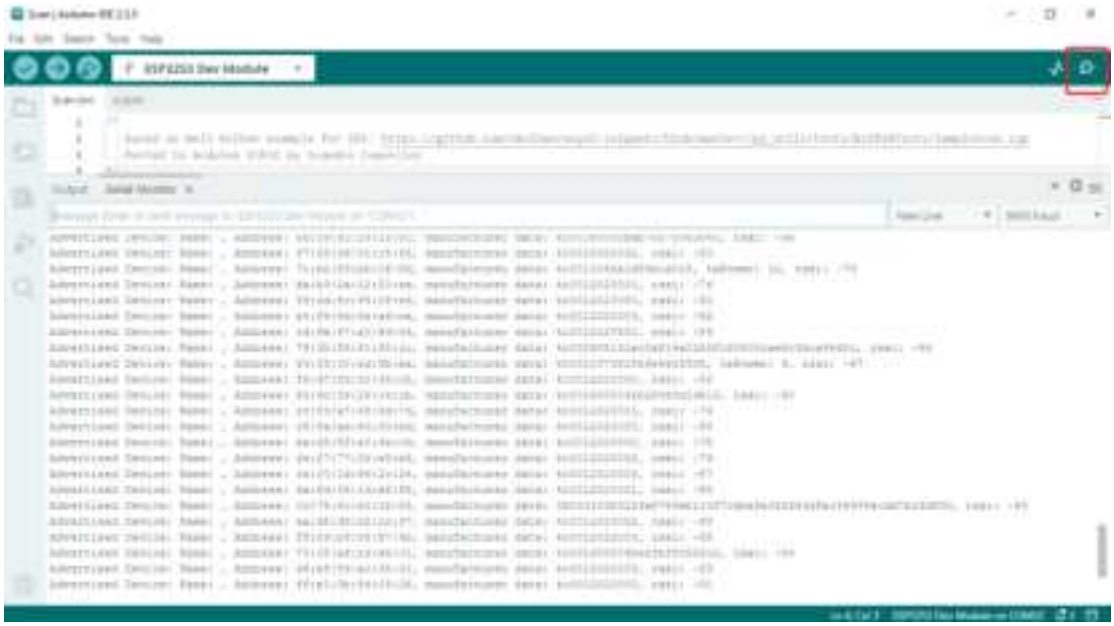


3.3 BLE Scan

Select the example program "Examples" → "BLE" → "Scan", choose the port corresponding to your device, and click the compile and upload button in the top-left corner.
After uploading is complete, open the Serial Monitor to view BLE scan information.


sketch_jun6a | Arduino IDE 2.3.3
File Edit Sketch Tools Help
New Sketch Ctrl+N
New Cloud Sketch Alt+Ctrl+N
Open... Ctrl+O
Open Recent
Sketchbook
Examples
Close Ctrl+W
Save Ctrl+S
Save As... Ctrl+Shift+S
Preferences... Ctrl+Comma
Advanced
Quit Ctrl+Q
I2C Display
OLEDImage
OSUSB
ESP32 Starter Kit_BasicKit
ESP32 Arduino IDE
Examples for ESP32S3 Dev Module
ArduinoOTA
BLE
BluetoothSerial
DNSServer
EEPROM
ESP Insights
ESP FlashMaker
ESP_I2S
ESP_NOW
ESP_SR
ESPFS
ESPFS2 Async UDP
ESPmDNS
Ethernet
Preferences
Firmata
HTTPClient
HTTPUpdate
HTTPUpdateServer
Keyboard
Beacon_Scanner
BLE3_extended_scan
BLE3_multi_advertising
BLE3_periodic_advertising
BLE3_periodic_sync
Client
EddystoneTLM_Beacon
EddystoneURL_Beacon
Beacon
Iot_Ally
Scan
Server
Server_multiconnect
UART
Write


Scan | Arduino IDE 2.3.3
File Edit Sketch Tools Help
ESP32S3 Dev Module
Output
Leaving...
Hard resetting via RTS pin...



4. FCC Warning

FCC Caution:

Any Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

IMPORTANT NOTE:

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

— Reorient or relocate the receiving antenna. — Increase the separation between the equipment and receiver. — Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.

— Consult the dealer or an experienced radio/TV technician for help.

FCC Radiation Exposure Statement: The device has been evaluated to meet general RF exposure requirement. The device can be used in portable exposure condition without restriction.